



Biological Responses in Terrestrial Systems

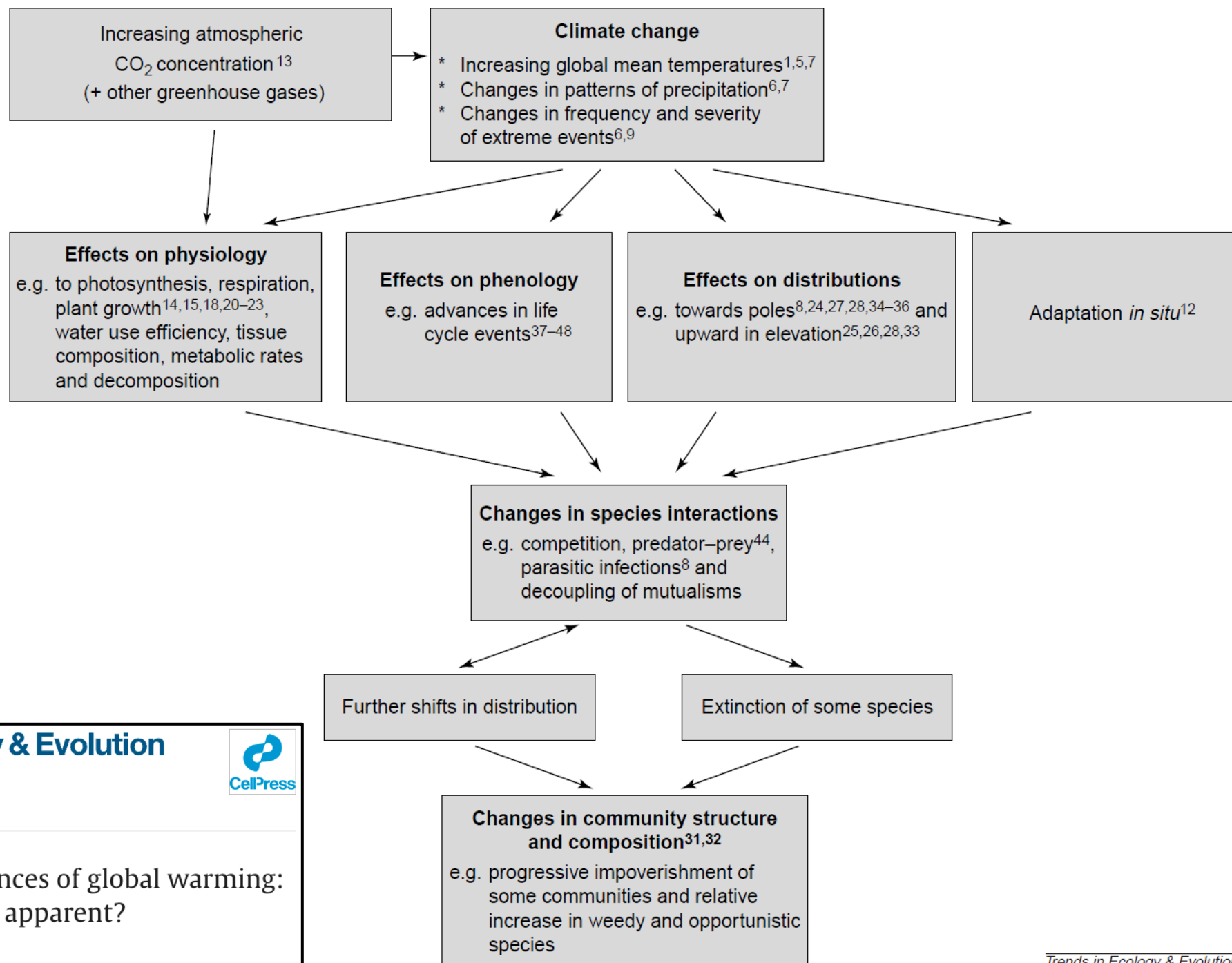
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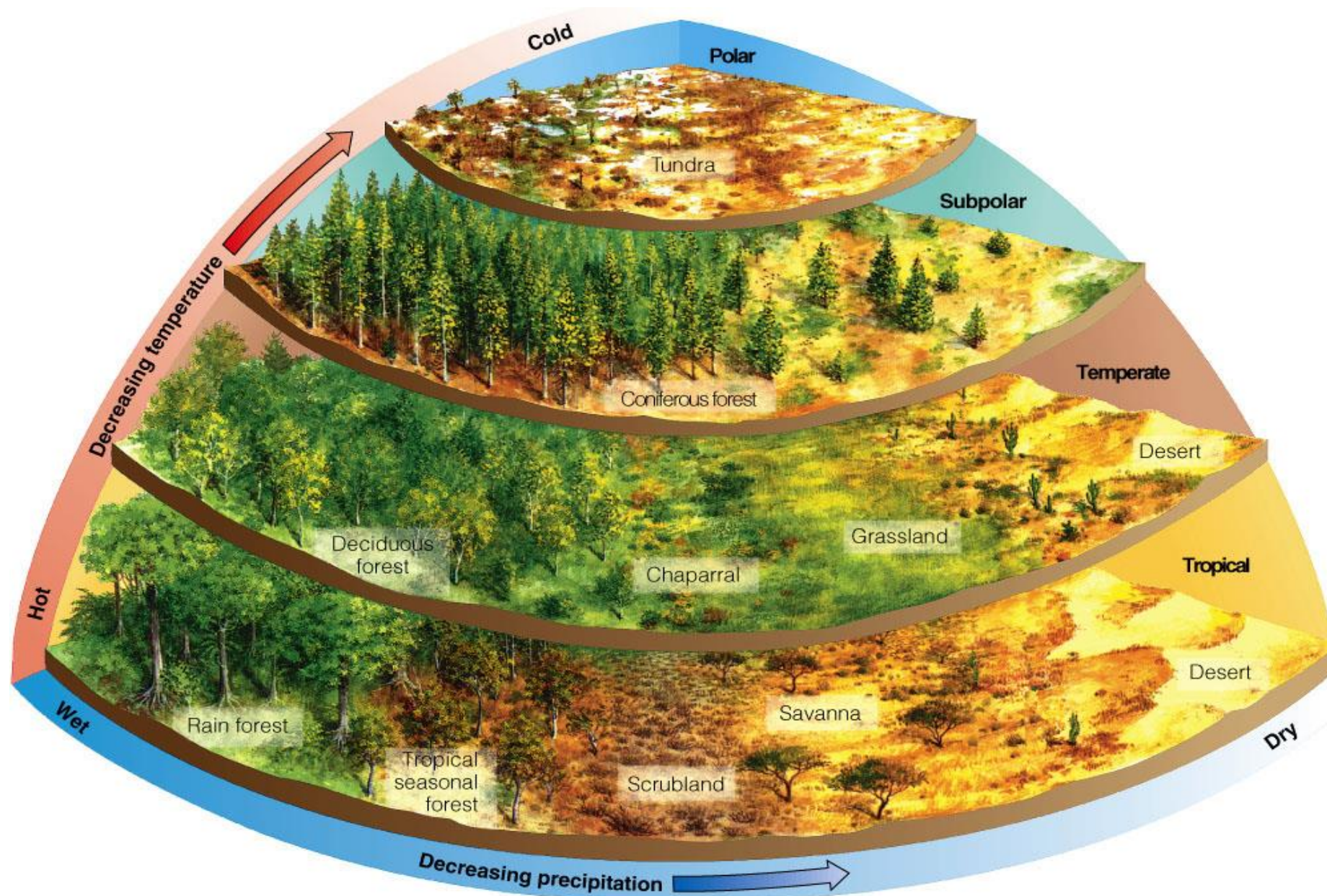
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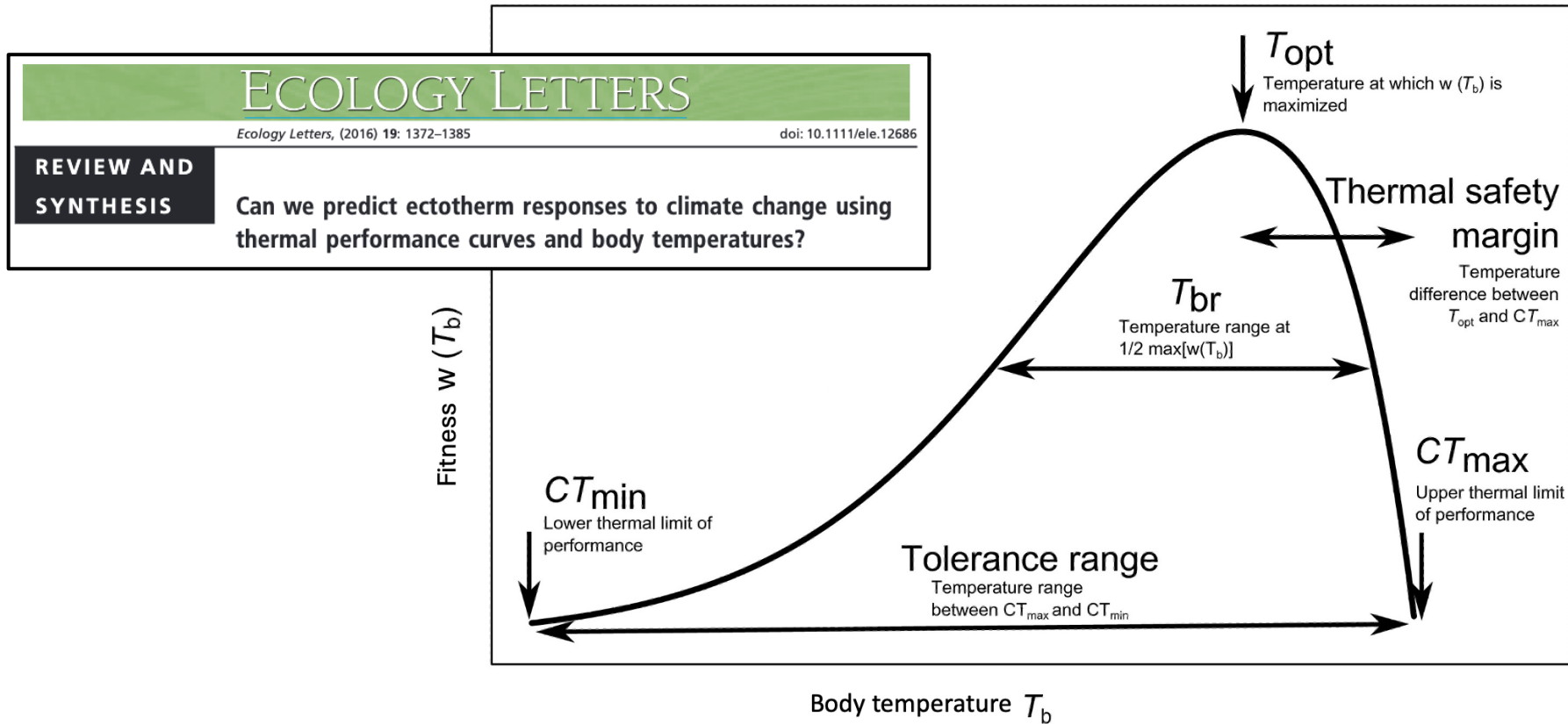
BIOMES AND SHIFTS



EFFECTS ON PHYSIOLOGY

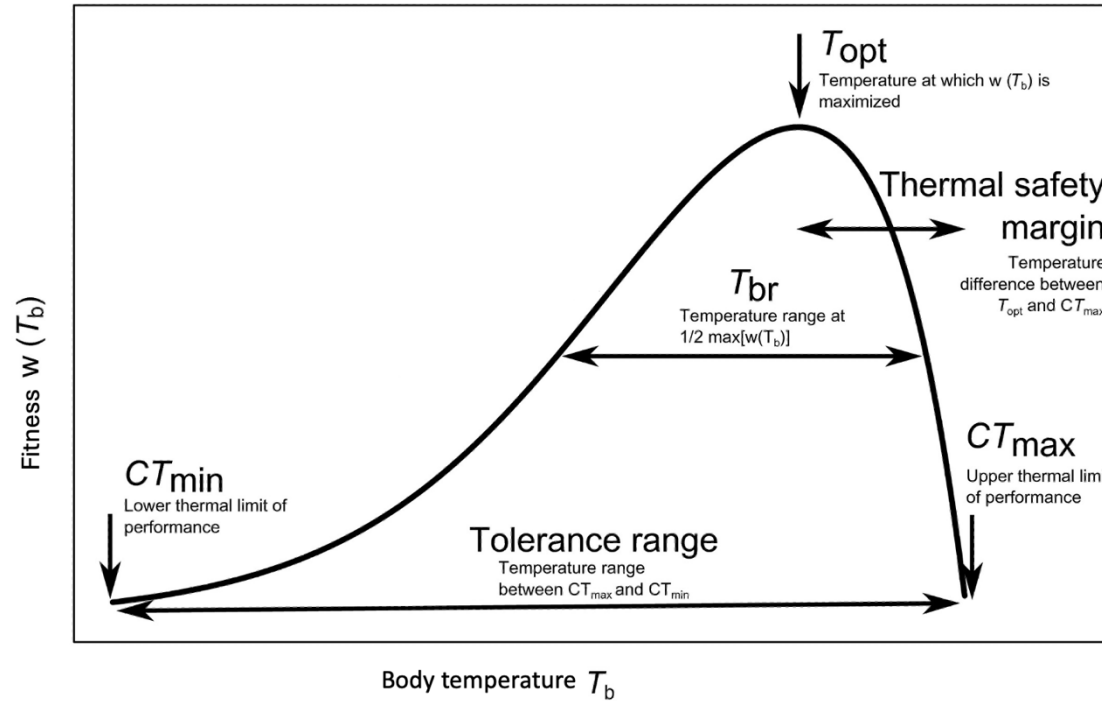
- Responses and tolerance to changing climatic conditions are determined by taxonomic and physiological context
- Each species has a **thermal tolerance range**—a specific band of temperatures within which it can survive, grow, reproduce, and perform a physiological function overall
- As temperature increases due to climate change, regions that were once within a species' comfort zone may become **too hot**, leading to **thermal stress**
- Organisms exposed to temperatures above their optimum:
 - Experience reduced reproductive success
 - Show weakened immune responses
 - Experience changes in respiration
 - Suffer higher mortality rates

EFFECTS ON PHYSIOLOGY



- Relationship between an ectotherm's body temperature and its biological performance (e.g., locomotion speed, growth rate, developmental rate, or metabolic activity)
- Essential for predicting how environmental temperature changes affect animal survival and fitness
 - Because ectotherms rely on external thermal sources to regulate their body temperature

EFFECTS ON PHYSIOLOGY



CT_{min} : The minimum body temperature at which an organism cannot maintain performance

CT_{max} : The maximum body temperature at which an organism cannot maintain performance

T_{opt} : The optimal temperature at which the organism's performance or fitness is maximized

T_{br} : Thermal breadth, the range of temperatures over which fitness remains above 50% of maximum

Thermal Safety Margin : Difference between the CT_{max} and the T_{opt} , indicating buffer against warming

Tolerance Range : The range between CT_{max} and CT_{min} where the organism can survive and perform

EFFECTS ON PHYSIOLOGY

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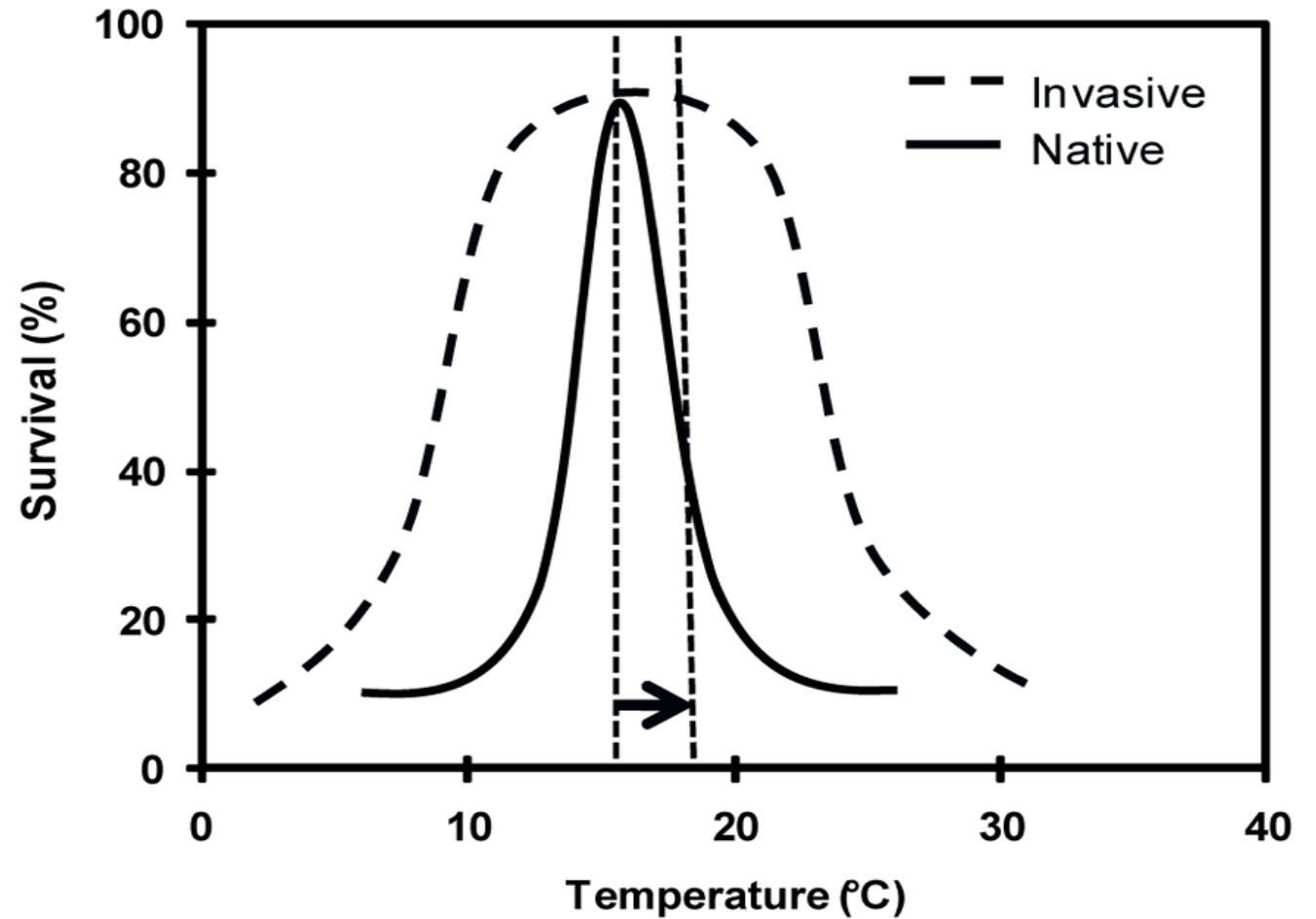
REVIEW ARTICLE

A peer-reviewed open-access journal
NeoBiota
Advancing research on alien species and biological invasions

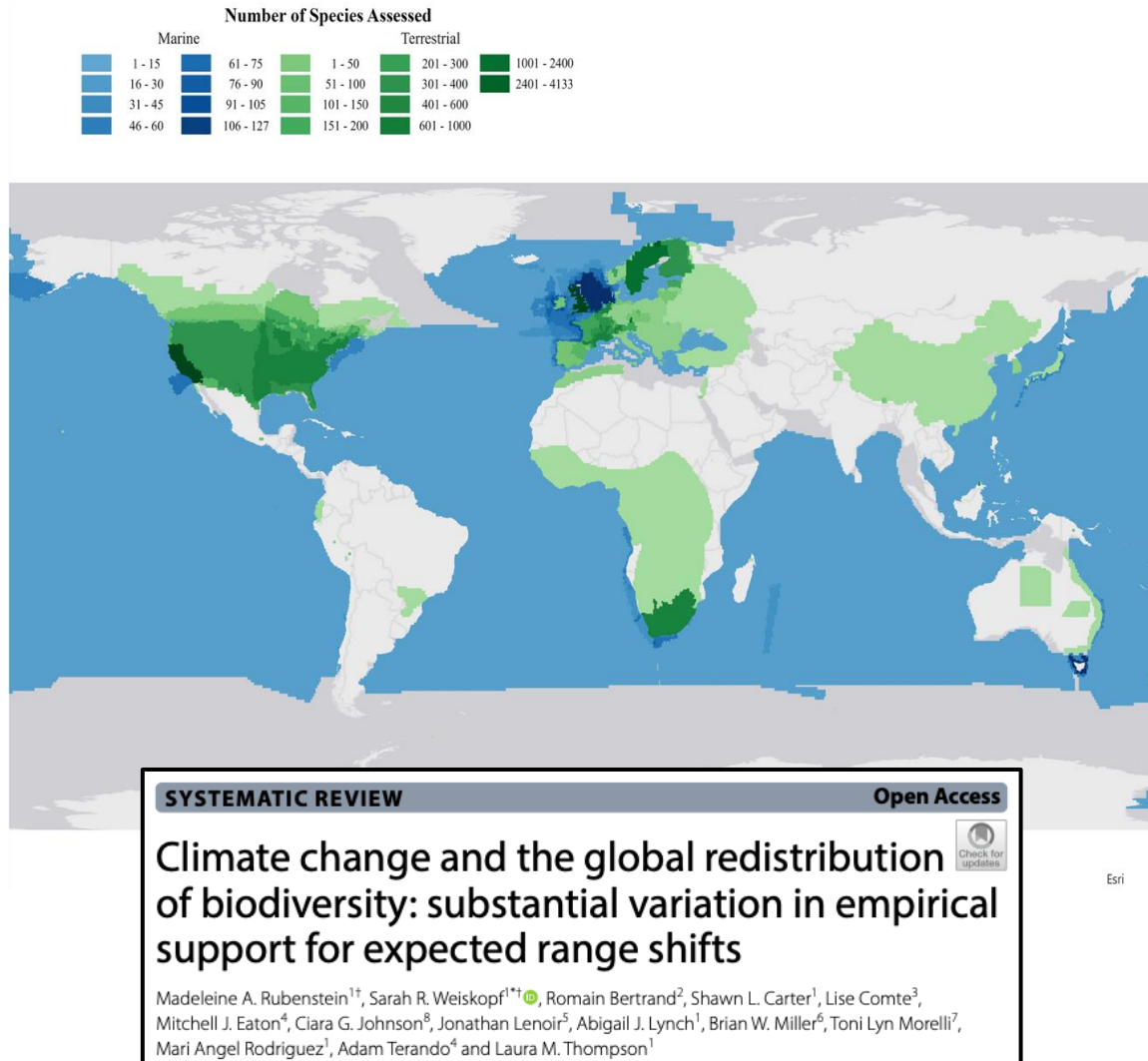
Double trouble: the implications of climate change for biological invasions

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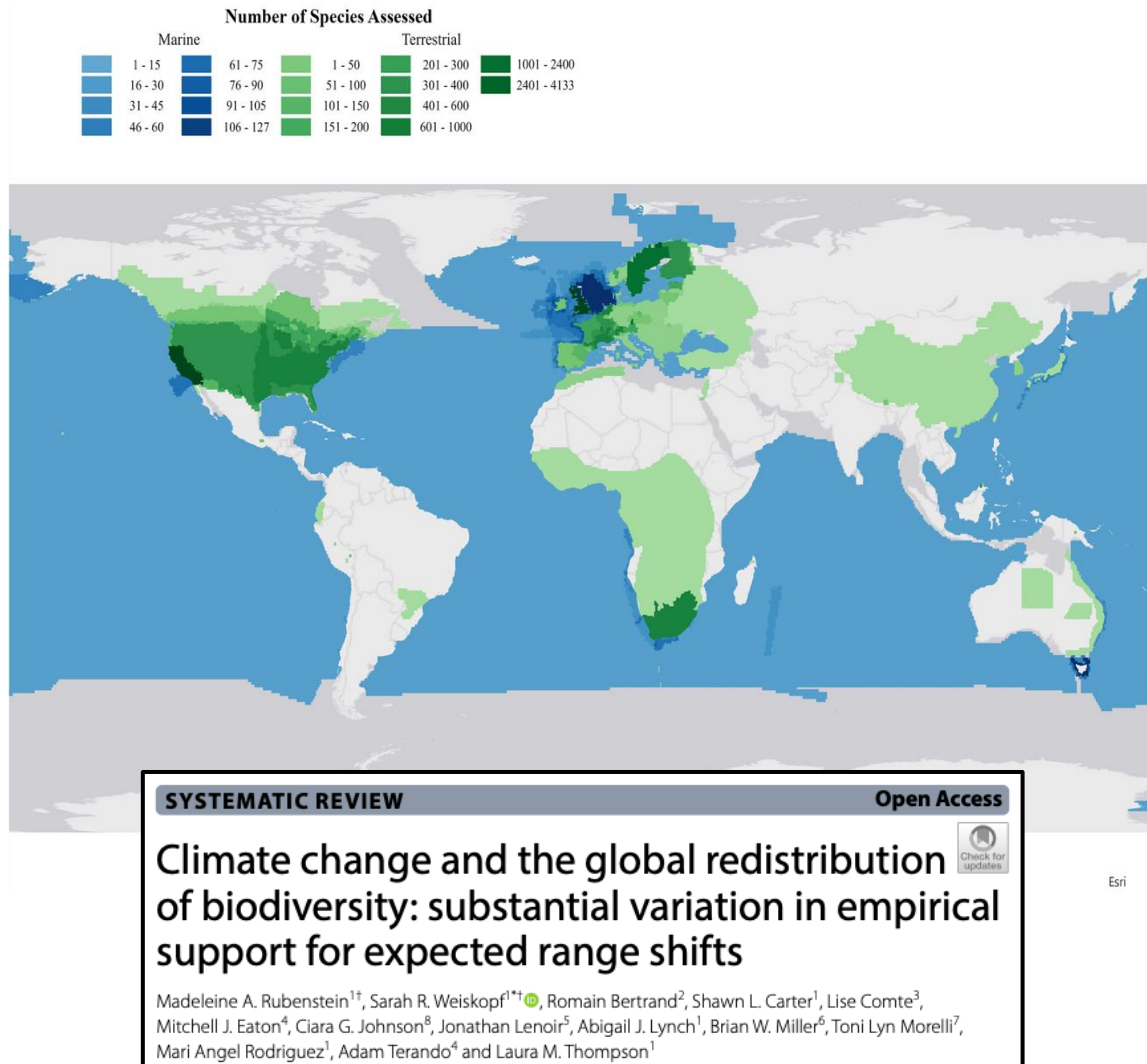
SPECIES DISTRIBUTION SHIFTS



Range Contractions and Expansions

- **Contractions:** As suitable habitats shrink due to climate change, some species may experience range contractions, leading to their extinction in certain areas
- **Expansions:** Some species may expand their ranges as they find new, more suitable habitats in areas that are becoming warmer or more moist

SPECIES DISTRIBUTION SHIFTS

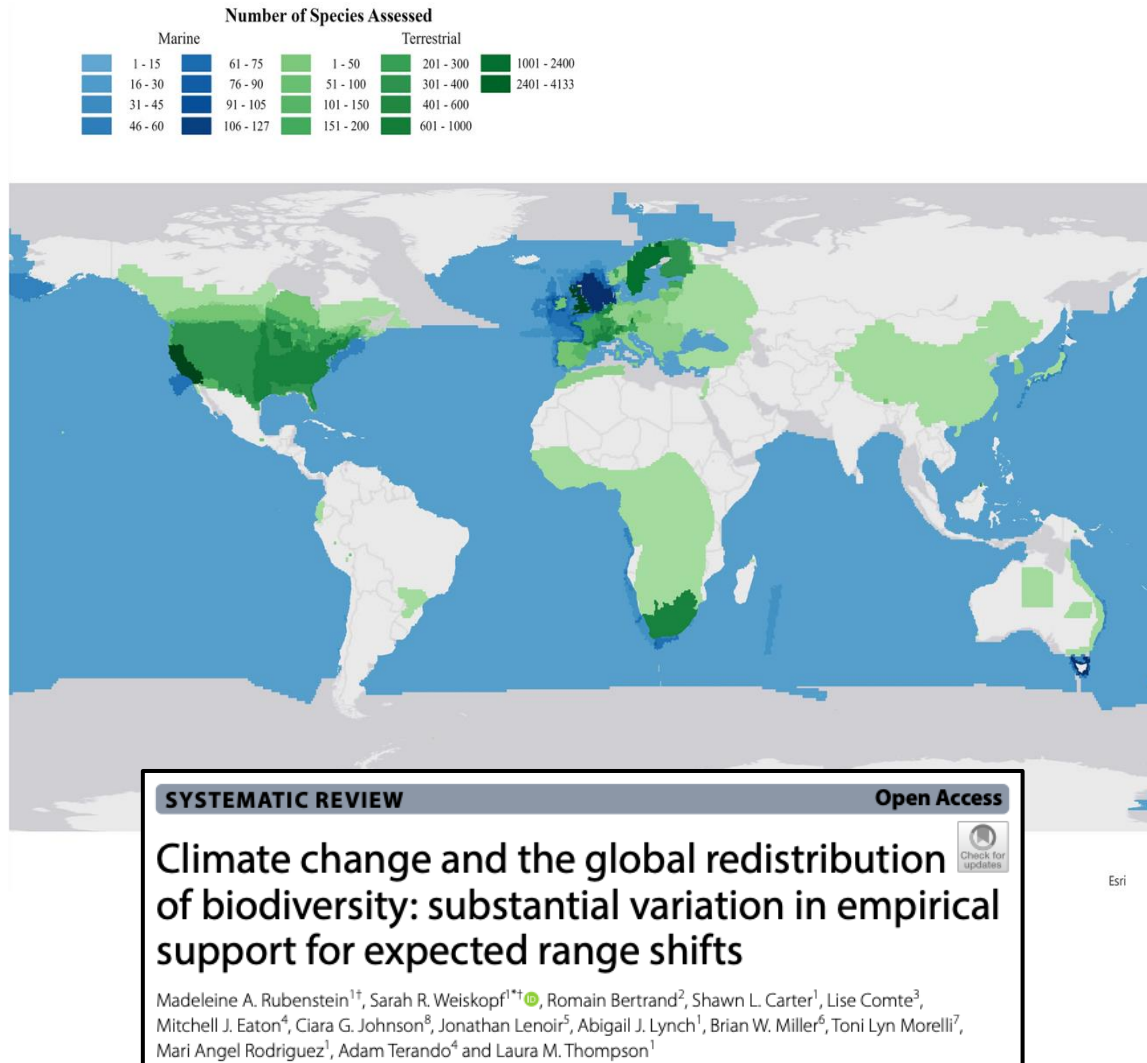


Range Shifts

- **Poleward Shifts:** As temperatures rise, many species are expected to shift their ranges towards cooler regions, such as higher latitudes or altitudes
- **Elevation Shifts:** Species may move to higher elevations to find cooler temperatures, particularly in areas with steep climate gradients, like mountain ranges

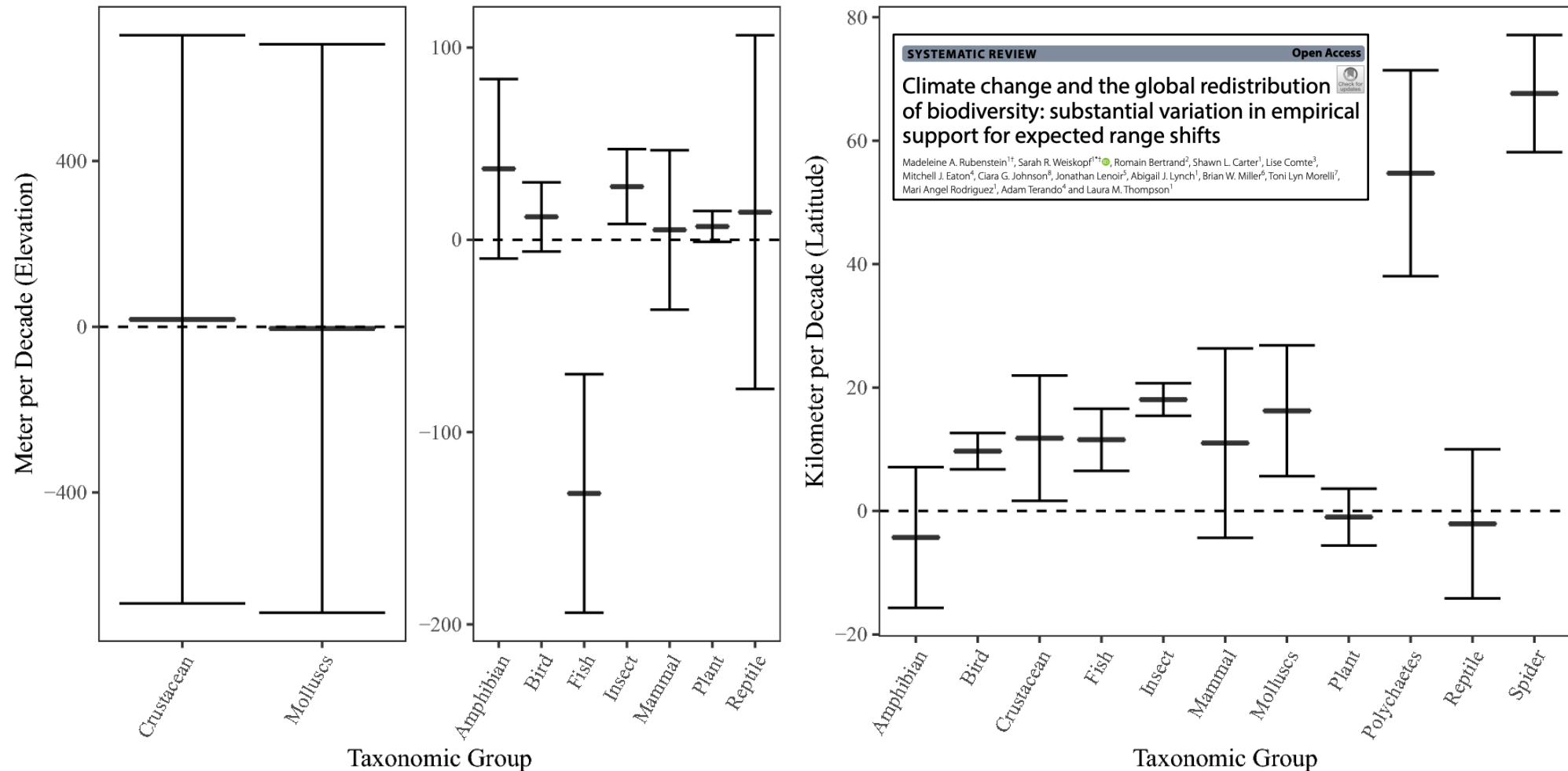
SPECIES DISTRIBUTION SHIFTS

Factors Influencing Range Shifts



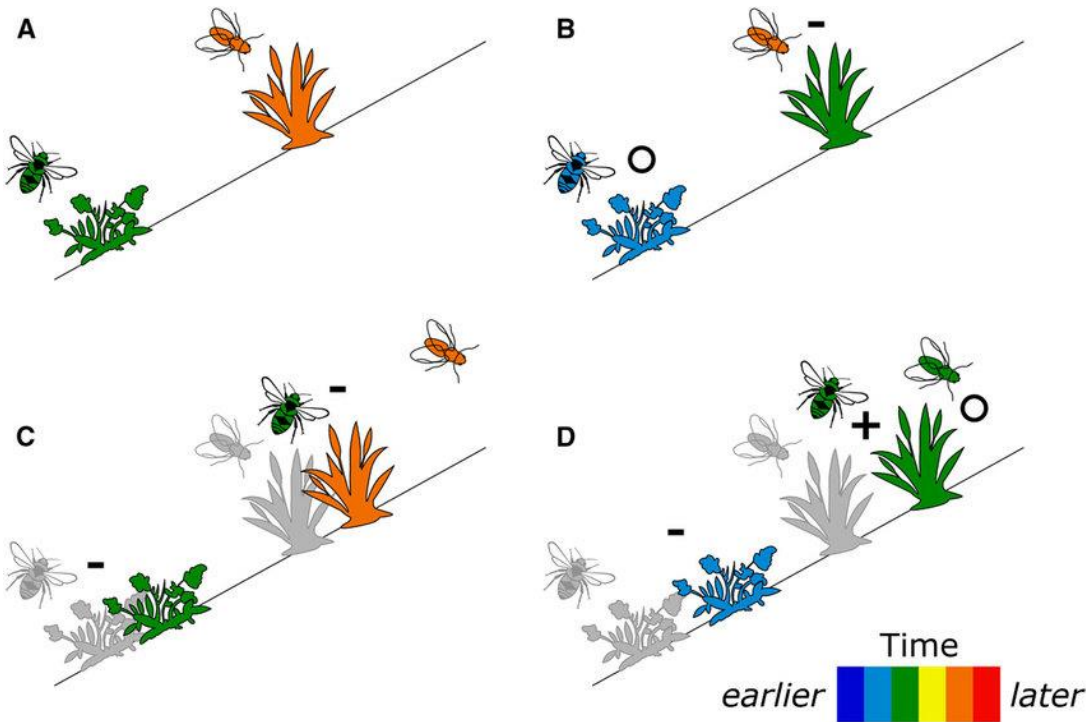
- **Temperature and Precipitation:** Changes in these factors are major drivers of range shifts, as species have specific temperature and precipitation tolerances
- **Habitat Fragmentation and Loss:** Can hinder species' ability to shift their ranges, as it reduces the availability of suitable areas for dispersal and colonization
- **Species-Specific Responses:** Different species respond to climate change differently, with some shifting ranges quickly while others may have more limited responses

SPECIES DISTRIBUTION SHIFTS



- **Elevational shifts:** amphibians, insects shifted upslope, whereas fish shifted downslope/greater depths
- **Latitudinal shifts:** in birds, insects, fish, crustaceans, polychaetes, and spiders

EFFECTS ON PHENOLOGY



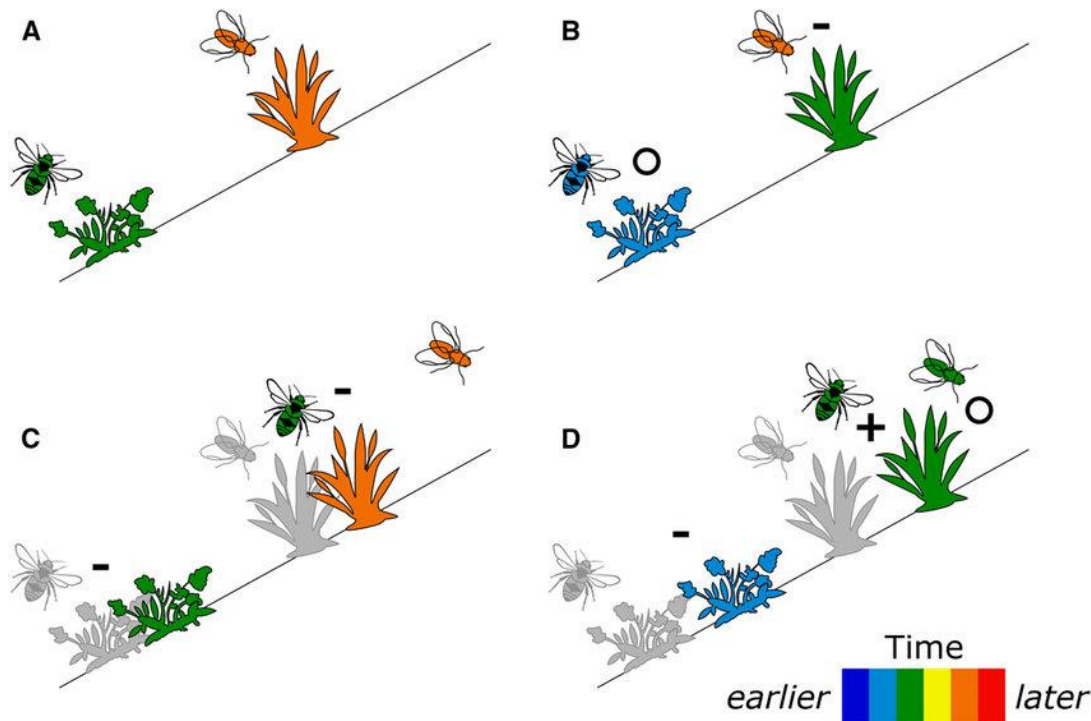
PLANT-POLLINATOR INTERACTIONS UNDER CLIMATE CHANGE: THE USE OF SPATIAL AND TEMPORAL TRANSPLANTS¹

EVA M. MORTON^{2,3,4} AND NICOLE E. RAFFERTY^{2,3,5}

- Climate change is causing widespread phenological shifts
 - the changes in the timing of life cycle events in plants and animals, with potential impacts on biodiversity and ecosystem functioning
- These shifts, often involving earlier timing of events like flowering and migration, are linked to rising temperatures, changes in rainfall patterns, and increased frequency of extreme weather events
- Unequal shifts across different species and trophic levels can lead to mismatches between species and their food sources, potentially disrupting ecological interactions and impacting species populations

EFFECTS ON PHENOLOGY

Plant-Pollinator Interaction Experiment



**PLANT-POLLINATOR INTERACTIONS UNDER CLIMATE CHANGE:
THE USE OF SPATIAL AND TEMPORAL TRANSPLANTS¹**

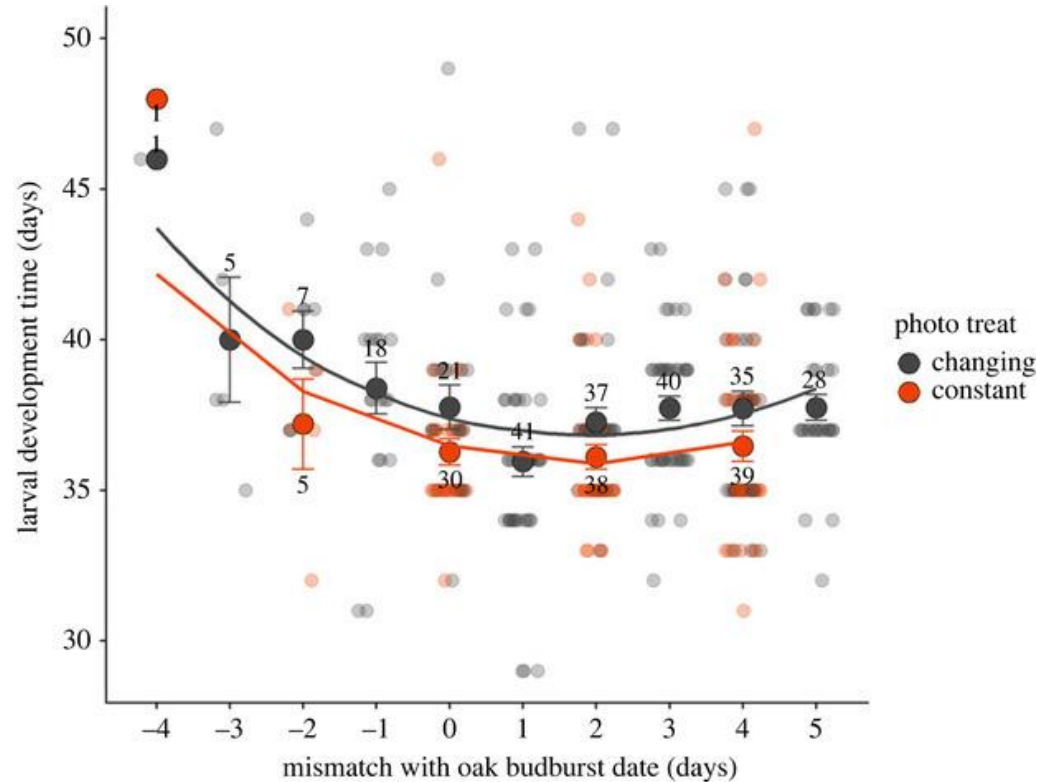
EVA M. MORTON^{2,3,4} AND NICOLE E. RAFFERTY^{2,3,5}

- A) Positions occupied by populations pre-climate change (historical baseline)
- (B) Phenological/temporal shifts alone
- (C) Distributional/spatial shifts alone
- (D) Joint phenological/temporal and distributional/spatial shifts
- (B–D) Each panel illustrates possible outcomes of shifts:
 - (i) maintenance of historical interactions (indicated by O)
 - (ii) loss of historical interactions (indicated by –)
 - (iii) gain of novel interactions (indicated by +)

EFFECTS ON PHENOLOGY

CASE STUDY

Effects of phenological mismatch on larval development time



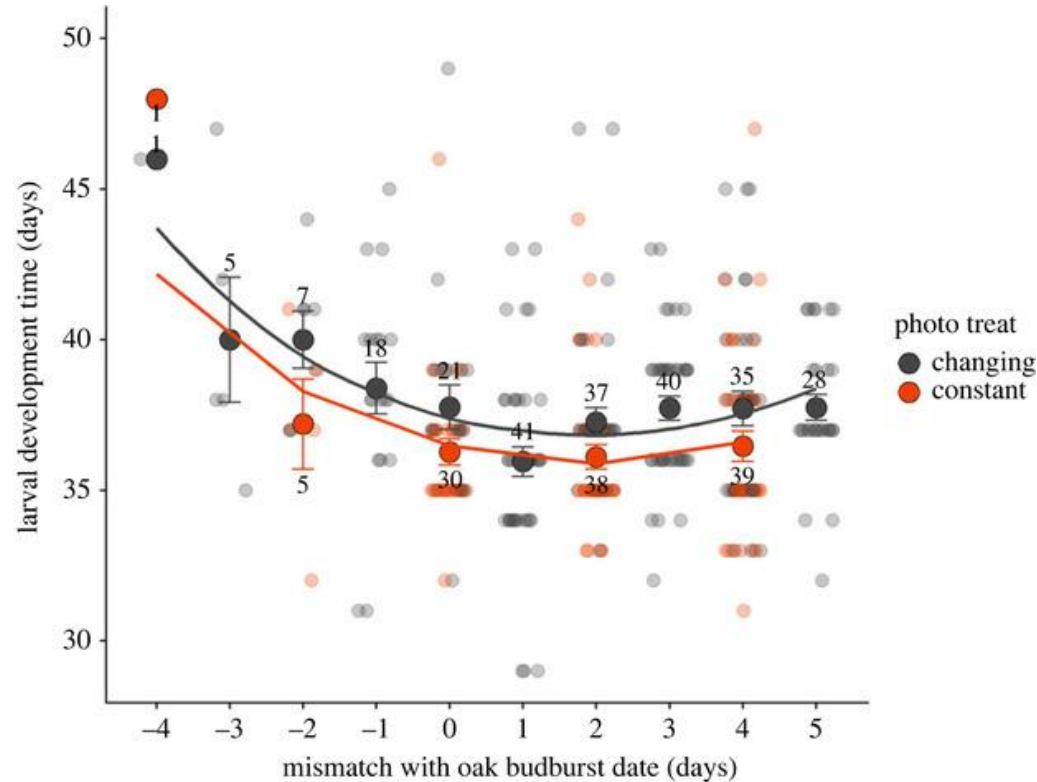
Phenological mismatch affects individual fitness and population growth in the winter moth

Natalie E. van Dis^{1,2}, Geert-Jan Sieperda¹, Vidisha Bansal^{1,2}, Bart van Lith¹, Bregje Wertheim² and Marcel E. Visser^{1,2}

- **How long caterpillars (larvae) take to grow** based on how well their hatching time syncs up with when oak trees open their leaves (**oak budburst**)
- **The Goal:** Caterpillars need fresh, soft oak leaves to eat
- **The Problem:** If caterpillars hatch too early or too late relative to leaf opening, their food quality changes, which affects their growth speed

EFFECTS ON PHENOLOGY

Timing Matters



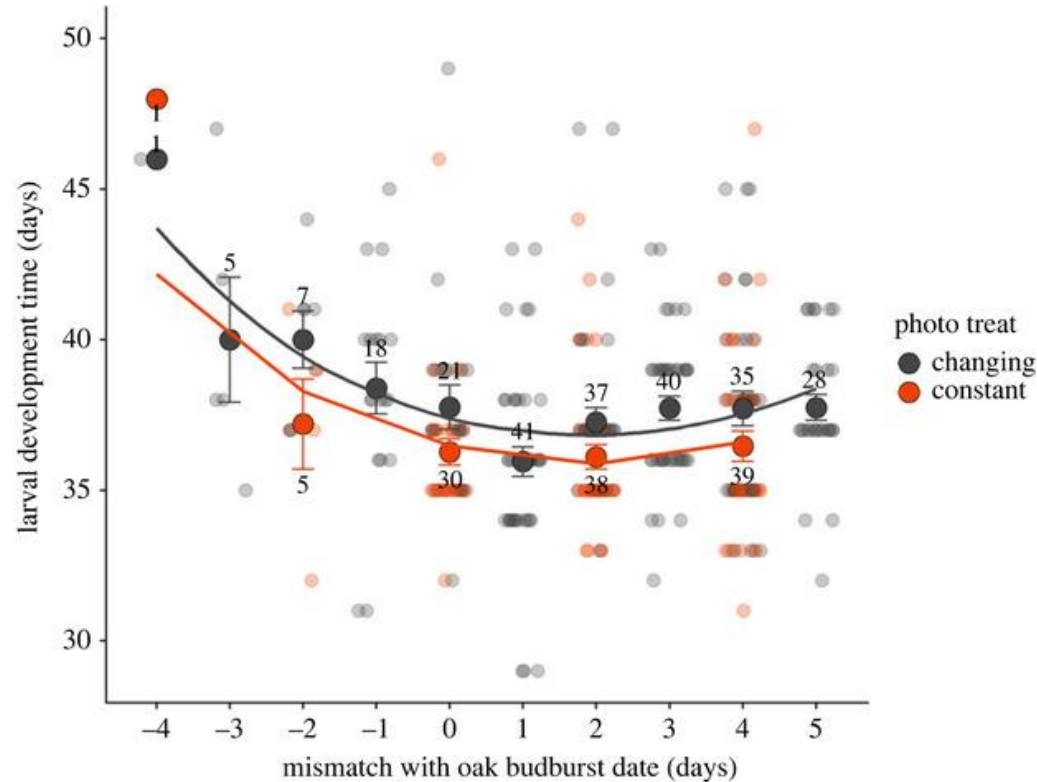
- **Perfect timing (0 to +2 days):** When larvae hatch around the exact time oak leaves open, they develop the **fastest** (taking around **36–37 days**)
- **Hatching too early:** If larvae hatch before the leaves open (e.g., -3 or -4 days early), their growth slows down drastically, taking up to **40–48 days** to mature

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EFFECTS ON PHENOLOGY

Daylight Conditions



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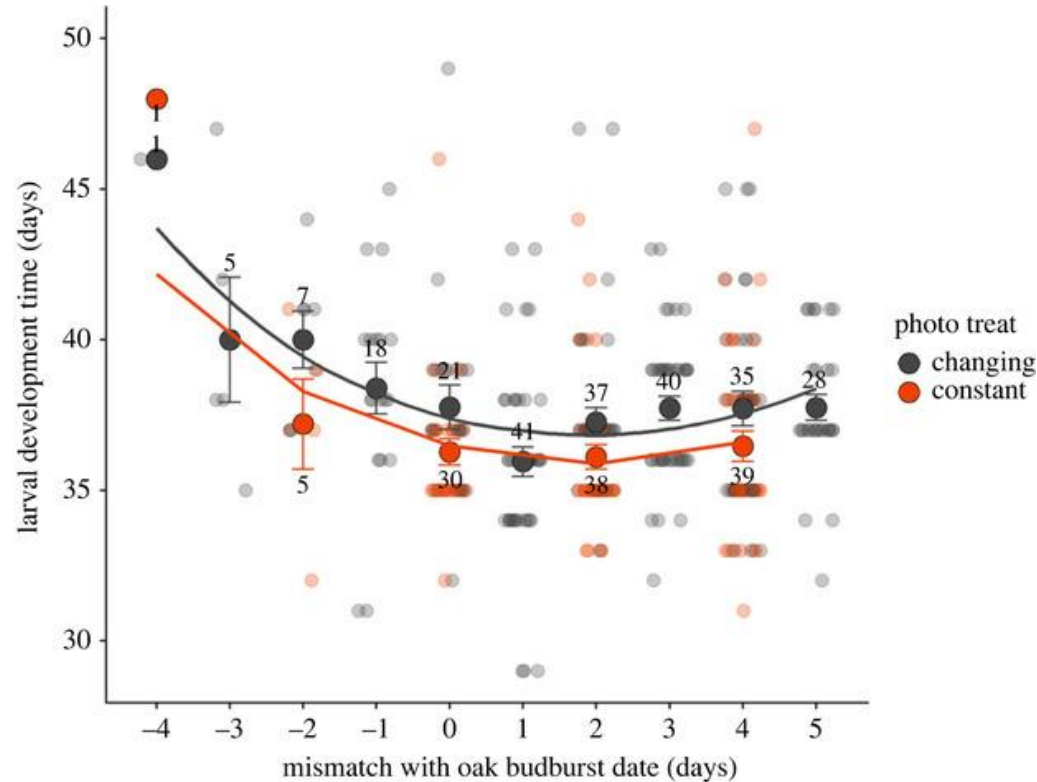
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- **Black Dots/Line ("changing"):** Larvae exposed to natural, changing daylight patterns (e.g., days getting longer as spring progresses)
- **Orange Dots/Line ("constant"):** Larvae exposed to a constant amount of light each day
- **Result:** Under almost all timing conditions, larvae in the **constant light group** grew slightly faster (took fewer days) than those experiencing **changing daylight**

EFFECTS ON PHENOLOGY

Conclusion

- Caterpillars grow fastest when their hatching perfectly aligns with the opening of fresh oak leaves
- Temporal shifts in hatching (e.g. hatching too early) significantly delays their growth
- Light conditions also play a role, with constant light slightly speeding up development compared to changing daylight
- This mismatch disrupts food availability for birds like the **Pied Flycatcher** because they feed on these caterpillars



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THANK YOU